Courier Management System (CMS)

User Story:

Courier and delivery organizations face difficulties in handling shipment and delivery activities when their operations depend on manual methods. Maintaining customer details, shipment records, delivery schedules, and payment information separately can cause errors and delays. Tracking parcels and updating delivery status manually also reduces transparency for customers. Assigning delivery agents without a proper system may lead to workload imbalance and scheduling problems. In addition, limited reporting facilities make it difficult for managers to monitor daily operations and identify areas for improvement. Managing information across multiple branches can also create communication and coordination challenges. Therefore, an integrated Courier Management System is needed to simplify operations, improve tracking, reduce manual work, and provide better service to customers.

1. INTRODUCTION

1.1 Project Overview :

Courier and logistics organizations often face difficulties in managing customer details, shipments, delivery schedules, and parcel tracking through manual methods. Using separate records and traditional processes can cause data errors, delays, and communication problems between staff, delivery agents, and customers.

The Courier Management System (CMS) is a Salesforce-based cloud application developed to organize these activities through a centralized platform. The system allows users to manage customer information, register shipments, assign delivery agents, update delivery status, maintain billing details, and generate reports.

By maintaining courier information in one system, the CMS reduces manual work and improves the accuracy and accessibility of records. It also helps administrators monitor shipment progress and delivery activities more effectively. Overall, the system supports faster operations, better coordination, improved shipment visibility, and enhanced customer service.

1.2 Purpose :

The main purpose of the Courier Management System (CMS) is to provide a centralized digital platform for managing courier and delivery activities efficiently. The system is designed to reduce manual paperwork and simplify the handling of customer details, shipment records, delivery schedules, and payment information.

The CMS enables users to create and manage shipments, assign delivery agents, update parcel status, and maintain accurate delivery records. It also helps administrators monitor ongoing and completed deliveries through organized records and reports.

Using Salesforce as the cloud platform, the system provides a structured environment for storing and accessing courier information. Overall, the project aims to reduce operational errors, improve delivery coordination, provide better shipment visibility, and enhance the quality of service offered to customers.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

2 IDEATION PHASE

2.2 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnected	stressed
PS-2	Customer	track parcels	updates are delayed	no centralized system	uncertain



Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



2.2 Customer Understanding

- **Hear:** Customers ask about parcel location, delivery time, and shipment status.
- **See:** Different shipment records, delivery schedules, and status information.
- **Say & Do:** Customers provide shipment details and check delivery updates through the system.
- **Pain:** Unclear delivery status, delayed notifications, and difficulty getting quick responses.
- **Gain:** Real-time shipment status, instant notifications, organized records, and easy tracking.

Short Version

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

2.3 Brainstorming & Idea Prioritization

Ideas Considered

- **Paper-Based Record System** – Maintaining shipment and customer details in physical files.
- **Excel-Based Tracking** – Recording parcel details and delivery status using spreadsheets.
- **Basic Web Application** – Developing a simple portal for shipment registration and tracking.
- **Cloud-Based Courier Management System** – Managing courier operations through a centralized Salesforce platform.

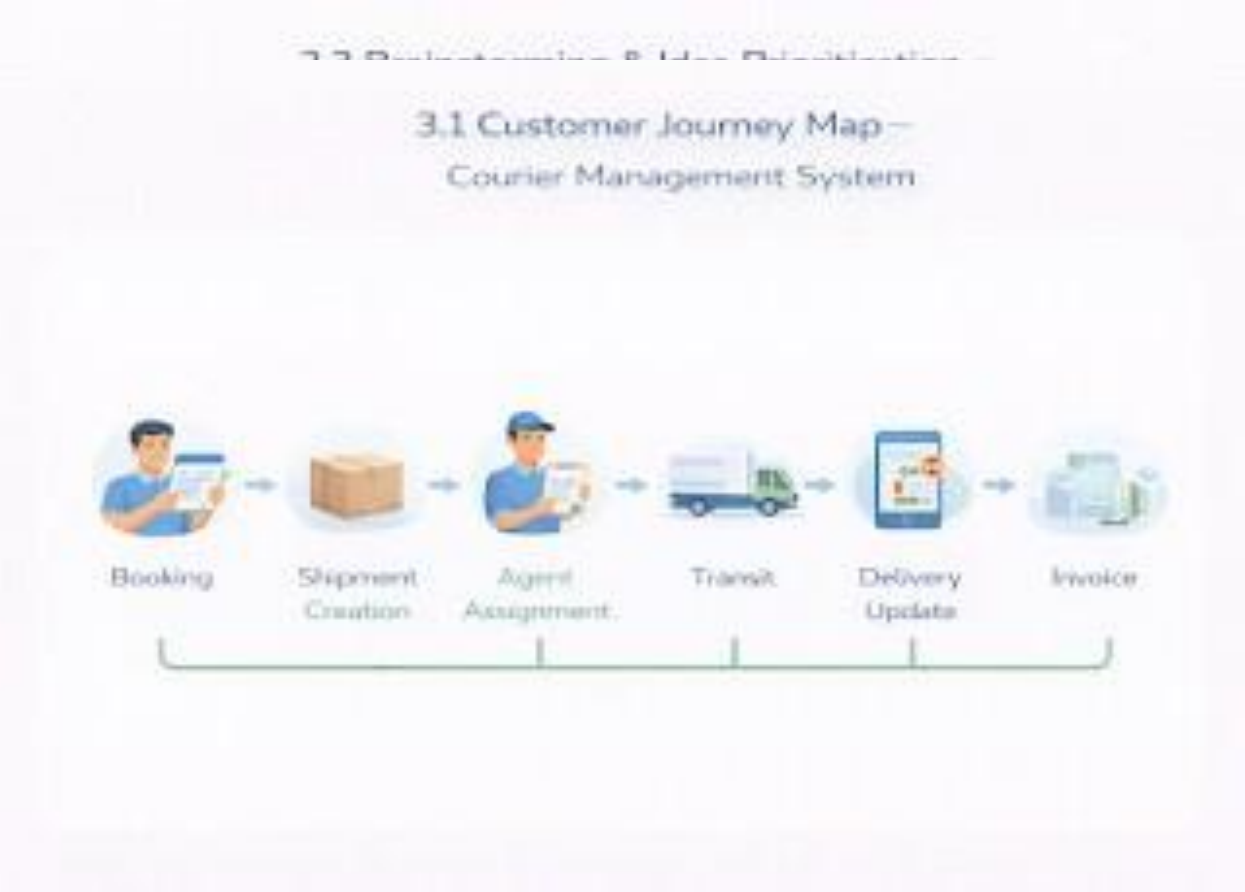
Final Selection

Cloud-Based Courier Management System using Salesforce was selected because it provides:

- **Centralized Data Managemnt** – Customer, shipment, agent, and invoice information can be maintained in one system.
- **Automated Workflows** – Shipment status and delivery activities can be updated automatically.
- **Easy Monitoring** – Managers can monitor shipments and delivery activities through reports and dashboards.
- **Secure Access** – User access can be controlled according to their roles and responsibilities.
- **Flexible Expansion** – The system can be extended as the courier business grows.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



• Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Non-Functional Requirements

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.



4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce CMS
Innovation	Automated tracking & billing

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

4.3 Solution Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management System.	3	High	Member1

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

			eꜰciently.			
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	1 2 Nov 2022		

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

6. Project Development Phase:

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Developing reports & dashboards
7. Setting up security with profiles, permission sets, and sharing
8. End-to-end application testing

Milestone 1: Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You're in the right place.

Welcome to the Loan Management System project!

Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you'll learn how Salesforce can streamline lending workflows—from borrower onboarding to loan approval, EMI tracking, and document verification.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :
3. First name & Last name
4. Email



1) First name & Last name
2) Email
3) Job Title: Developer
4) Company: College Name
5) Country: India

Click on sign me up after filling these.

Activity 2: Account Activation

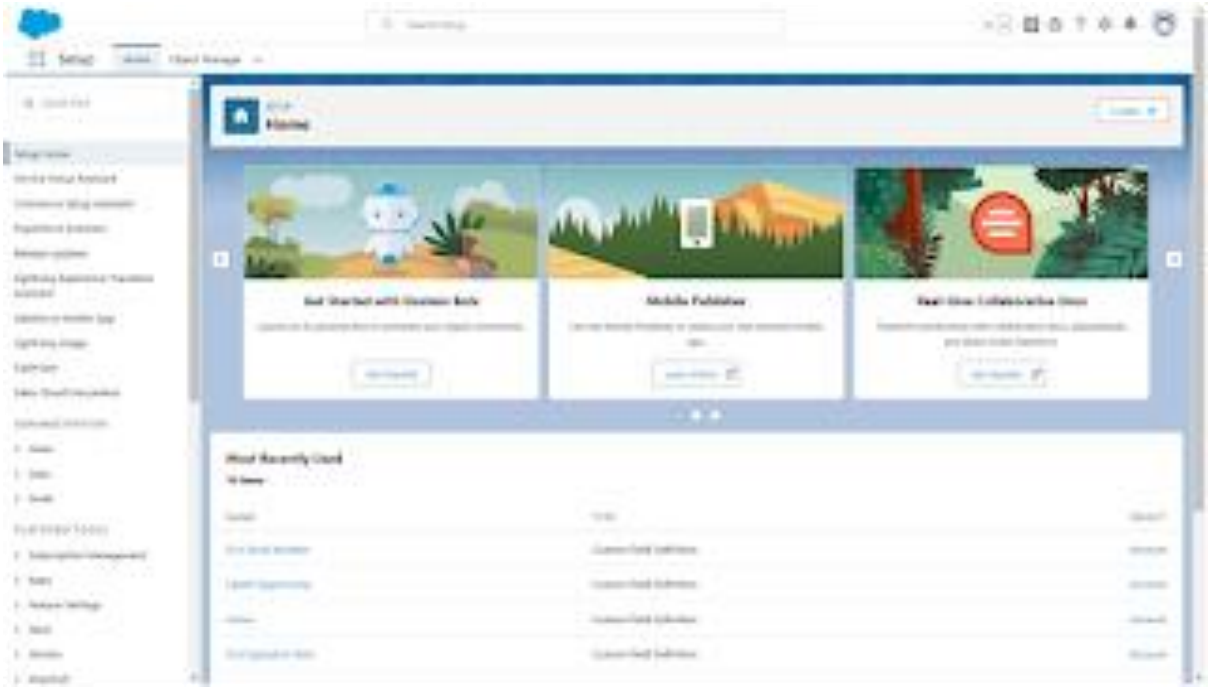
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.
4. Then you will redirect to your salesforce setup page.



Milestone 2- Object:

What is an object?

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

1. **Customer__c** – stores customer personal and contact information
2. **Shipment__c** – manages parcel details, source, destination, and tracking information
3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details
4. **Branch__c** – represents courier branch offices
5. **Delivery_Update__c** – stores status updates for each shipment
6. **Invoice__c** – manages billing details for delivered shipments

Relationships:

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

Delivery_Agent → Branch (Lookup)

Delivery_Update → Shipment (Master-Detail)

Invoice → Shipment (Lookup)

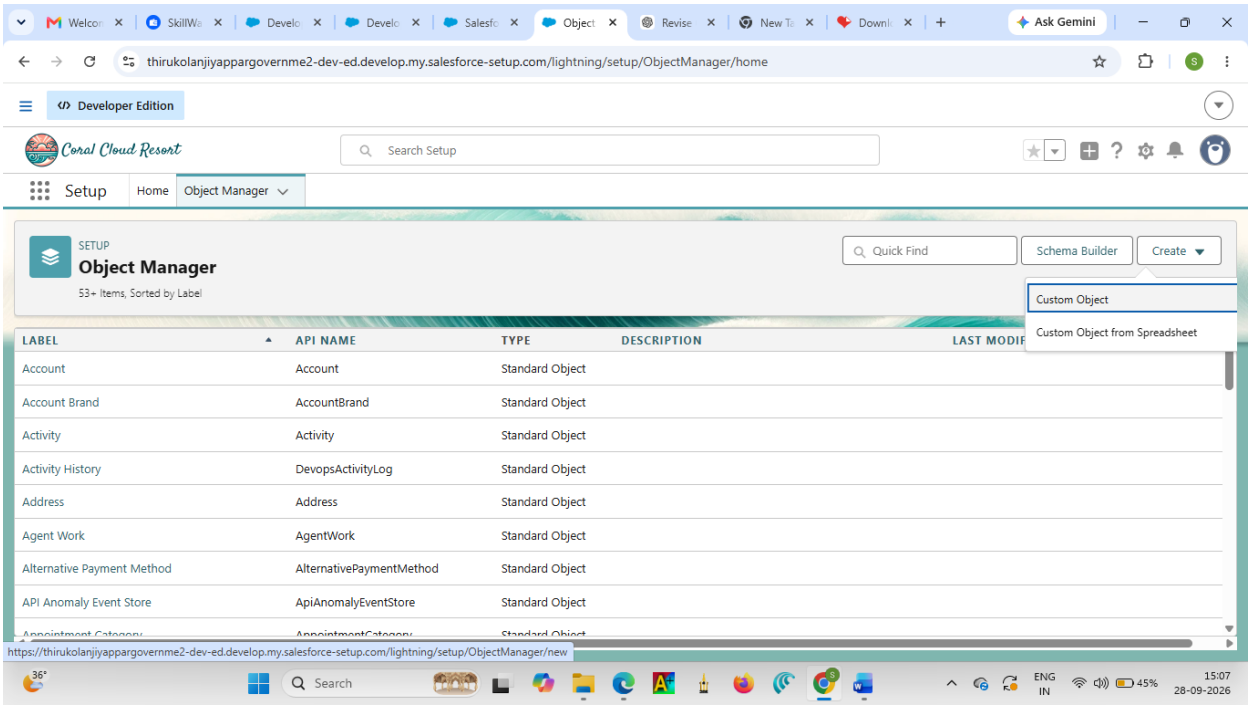
Steps to Implement :

Activity 1: Create Customer Object

1. Click Setup → Object Manager → Create → Custom Object

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



2 Enter the following details:

- Label: Customer
- Enable Allow Reports
- Enable Track Field History
- In Search Status check the box Allow Search
- In Object Creation Options check the box Add Notes and Attachments related list to default page layout

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

4. Click Save.



Activity 2: Create Shipment Object

1. Go to Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Shipment
- Plural Label: Shipments
- Record Name: Tracking Number (Auto Number)

3. Example Format: SHP-{00000}

Starting Number: 0

4. Enable:

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

2. Enter:

- Label: Delivery Agent
- Plural Label: Delivery Agents
- Record Name: Agent Name (Text)

3. Enable:

- Allow Reports
- Track Field History

4. Click Save

Activity 4: Create Branch Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Branch
- Plural Label: Branches
- Record Name: Branch Name (Text)

3. Enable:

- Allow Reports

4. Click Save.

Activity 5: Create Delivery_Update Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Delivery Update
- Plural Label: Delivery Updates
- Record Name: Update Number (Auto Number)

3. Example Format: UPD-{0000}

Enter Starting number : 0

4. Enable:

- Allow Reports
- Track Field History

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Label: Invoice
- Plural Label: Invoices
- Record Name: Invoice Number (Auto Number)

3. Example Format: INV-{0000}

Enter Starting number: 0

4. Enable:

- Allow Reports
- Track Field History

5. Click Save

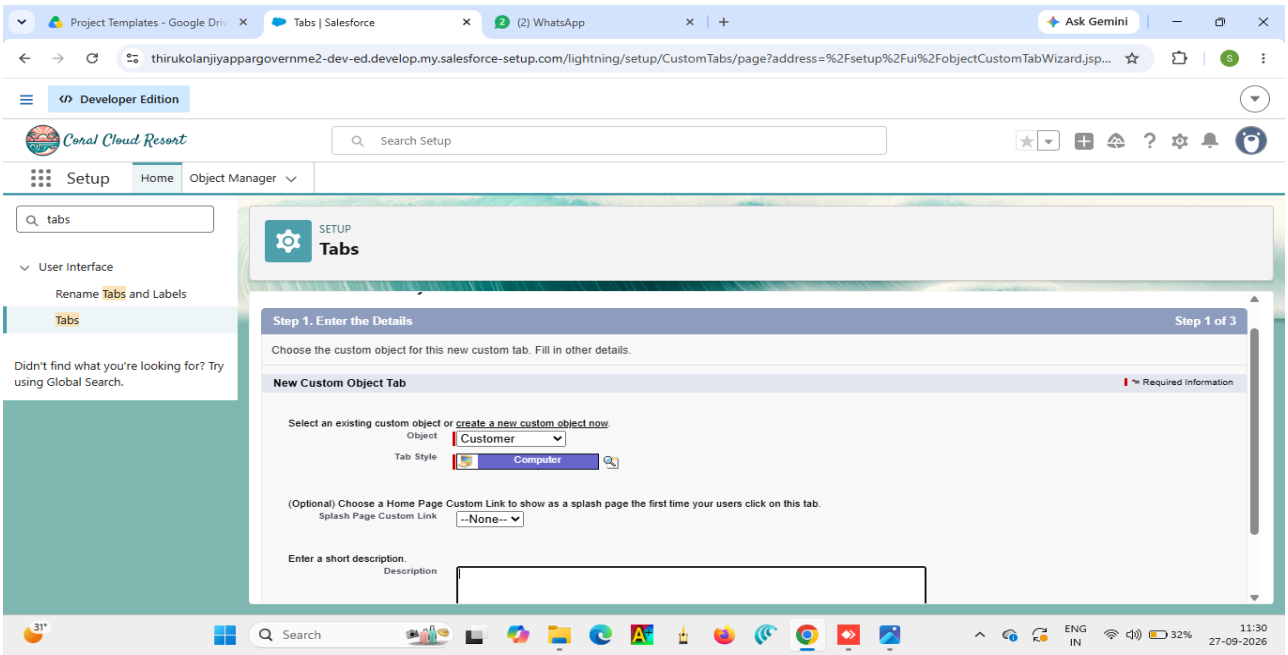
Milestone 3 - Tabs

Tab:

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1: Creating a Tab for Customer Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.



3. For Object, select Customer.
4. For Tab Style, choose any icon.
5. Leave all default settings as they are.

6. Click Next, Next, and Save.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

3. For Object, select Shipment.
4. Choose any Tab Style.
5. Keep defaults unchanged.
6. Click Next, Next, and Save.

Activity 3: Creating a Tab for Delivery Agent Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Agent.
4. Pick any Tab Style.
5. Leave default settings.
6. Click Next, Next, and Save.

Activity 4: Creating a Tab for Branch Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Branch.
4. Choose any Tab Style.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

Activity 5: Creating a Tab for Delivery Update Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Update.
4. Select any Tab Style.
5. Keep default configurations.
6. Click Next, Next, and Save.

Activity 6: Creating a Tab for Invoice Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Invoice.
4. Select any Tab Style.
5. Leave defaults unchanged.
6. Click Next, Next, and Save.

Milestone 4: Fields & Relationships

What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Email	Email__c	Email	Yes	Customer email
Phone	Phone__c	Phone	Yes	Customer contact number
Address	Address__c	Text Area	Yes	Customer address
Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer city
State	State__c	Text	No	Customer state
Pincode	Pincode__c	Text	No	Postal code
Is Active	Is_Active__c	Checkbox	No	Active status

1. Email (Email Field)

- Go to Setup → Object Manager → Customer__c
- Click Fields & Relationships → New
- Select Email → Next
- Field Label: Email
- Check the Required box
- Click Next → Next → Save

2. Phone (Phone Field)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Phone → Next

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

3. Address (Text Area)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Textarea → Next
- Field Label: Address
- Check Required
- Save

4. Customer Type (Picklist)

- Setup → Customer__c → Fields → New
- Picklist → Next
- Field Label: Customer Type
- Enter values:
    - Regular
    - Business
- Check Required
- Save

5. City (Text)

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Save

6. State (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **State**
- Save

7. Pincode (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **Pincode**
- Save

8. Is Active (Checkbox)

- Setup → Customer__c → Fields → New
- Checkbox → Next
- Field Label: **Is Active**
- Save

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

Field Name	API Name	Data Type	Required	Description
Tracking Number	Name	Auto Number	Yes	Unique ID
Customer	Customer__c	Lookup(Customer)	Yes	Related customer
Source Address	Source_Address__c	Text Area	Yes	Pickup location
Destination Address	Destination_Address__c	Text Area	Yes	Delivery location
Weight	Weight__c	Number	Yes	Parcel weight
Price	Price__c	Currency	Yes	Shipping charge
Status	Status__c	Picklist	Yes	Shipment status
Expected Date	Expected_Date__c	Date	Yes	Delivery estimate
Assigned Agent	Assigned_Agent__c	Lookup(Delivery_Agent)	No	Delivery agent
Shipment Type	Shipment_Type__c	Picklist	Yes	Domestic/Internatio

1. **Source Address (Text Area)**
   - Setup → Shipment__c → Fields → New
   - Textarea → Next
   - Field Label: **Source Address**
   - Required → Save
2. **Destination Address (Text Area)**

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Required → Save

3. Weight (Number)

- Setup → Shipment__c → Fields → New
- Number → Next
- Field Label: **Weight**
- Required → Save

4. Price (Currency)

- Setup → Shipment__c → Fields → New
- Currency → Next
- Field Label: **Price**
- Required → Save

5. Status (Picklist)

Values:

- Booked
- In Transit
- Out for Delivery

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

6. Expected Date (Date)

- Setup → Shipment__c → Fields → New
- Date → Next
- Field Label: **Expected Date**
- Required → Save

7. Shipment Type (Picklist)

Values:

- Domestic
- International

Required → Save

Activity 3: Creation of fields for Delivery_Agent Object

Field Name	API Name	Data Type	Required
Agent Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Branch	Branch__c	Lookup	Yes
Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Field Label: Email
- Required → Save

2. Phone

- Setup → Delivery_Agent__c → Fields → New
- Phone → Next
- Field Label: Phone
- Required → Save

3. Status (Picklist)

Values:

- Active
- Inactive
- On Leave

Required → Save

4. Joining Date

- Setup → Delivery_Agent__c → Fields → New
- Date → Next

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Field Name	API Name	Data Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

Create each field using:

- Setup → Branch__c → Fields → New
- Select respective Data Type
- Enter Field Label
- Save

(Repeat for all fields)

Activity 5: Creation of fields for Delivery_Update Object

Field Name	API Name	Data Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail	Yes
Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	DateTime	Yes
Remarks	Remarks__c	Text	No

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

- Delivered
- Delayed

Required → Save

Activity 6: Creation of fields for Invoice Object

Field Name	API Name	Data Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Payment Status Values

- Paid
- Unpaid
- Pending

Required → Save

Relationships :

Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Relationship Creation Steps :

Activity 1 – Shipment → Customer

- Setup → Shipment__c → Fields → New
- Lookup Relationship
- Related Object: **Customer__c**
- Field Label: **Customer**
- Save

Activity 2 – Shipment → Delivery Agent

- Setup → Shipment__c
- New → Lookup
- Object: **Delivery_Agent__c**
- Label: **Assigned Agent**
- Save

Activity 3 – Delivery Agent → Branch

- Setup → Delivery_Agent__c
- New → Lookup

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 4 – Delivery Update → Shipment

- Setup → Delivery_Update__c
- New → **Master-Detail**
- Related Object: **Shipment__c**
- Save

Activity 5 – Invoice → Shipment

- Setup → Invoice__c
- New → Lookup
- Object: **Shipment__c**
- Save

Milestone 5-Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

1. Validation Rule: Email Must Not Be Blank

Steps:

- Click the Gear icon → Setup
- Go to Object Manager
- Search and select Customer__c
- From the left menu, click Validation Rules
- Click New
- Rule Name: Email_Required
- Check Active

In Error Condition Formula, paste:

ISBLANK(Email__c)

- Click Check Syntax
- Error Message: Email is required.
- Error Location: Field → Email
- Click Save.

2. Validation Rule: Phone Must Be 10 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

LEN(Phone _ c) <> 10

- Check Syntax
- Error Message: **Phone number must be 10 digits.**
- Error Location: Field → Phone
- Save

3. Validation Rule: Pincode Must Be 6 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Pincode
- Active → Yes

Formula:

LEN(Pincode__ c) <> 6

- Error Message: **Pincode must be 6 digits.**
- Location: Field → Pincode
- Save

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

- Setup → Object Manager → Shipment__c
- Click Validation Rules → New
- Rule Name: Validate_Weight
- Active → Checked

Formula:

Weight _ c <= 0

- Check Syntax
- Error Message: **Weight must be greater than zero.**
- Location: Field → Weight
- Save

2. Validation Rule: Price Must Be Greater Than Zero

Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Price
- Active → Yes

Formula:

Price _ c <= 0

- Error Message: **Price must be greater than zero.**
- Location: Field → Price
- Save

3. Validation Rule: Expected Date Cannot Be in the Past

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Rule Name: Validate_Expected_Date

Formula:

Expected_Date__c < TODAY()

- Check Syntax
- Error: **Expected delivery date cannot be in the past.**
- Location: Field → Expected Date
- Save

Activity 3: Create Validation Rules on Delivery_Agent___c

1. Validation Rule: Email Must Be Valid

Steps:

- Setup → Object Manager → Delivery_Agent__c
- Validation Rules → New
- Rule Name: Validate_Agent_Email
- Active → Yes

Formula:

NOT(CONTAINS(Email__c, "@"))

- Error Message: **Enter a valid email address.**

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 4: Create Validation Rules on Branch__c

1. Validation Rule: Branch Code Must Not Be Blank

Steps:

- Setup → Object Manager → Branch_c
- Validation Rules → New
- Rule Name: Branch_Code_Required

Formula:

ISBLANK(Branch_Code___c)

- Error Message: **Branch Code is mandatory.**
- Location: Field → Branch Code
- Save

Activity 5: Create Validation Rules on Delivery_Update__c

1. Validation Rule: Remarks Required If Status is Delayed

Steps:

- Setup → Object Manager → Delivery_Update_c
- Validation Rules → New
- Rule Name: Remarks_Required_For_Delay

Formula:

```
AND(
 ISPICKVAL(Status_c, "Delayed"),
 ISBLANK(Remarks_c)
)
```

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Milestone 6 - Page Layouts :

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.
They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.
By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

Activity 1: Edit Page Layout for Customer__c

Steps:

- Setup → Object Manager → Customer__c
- Click Page Layouts
- Click the default layout (usually Customer Layout)

Sections to Add & Arrange:

- Open the default layout
- In the top palette, click “Section”
- Drag the Section and drop it anywhere on the layout

A pop-up appears → Fill details:

- Section Name: “Customer Information”
- Layout:

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Click OK

The new section will appear on the layout.

- Drop fields inside the section (it will highlight).
- Repeat the same for each field.

Section 1: Customer Information

- Customer Name
- Customer Type
- Email
- Phone
- Is Active

Section 2: Address Details

- Address
- City
- State
- Pincode

9. Edit the layout as needed

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 2: Edit Page Layout for Shipment__c

Steps:

- Setup → Object Manager → Shipment__c
- Click Page Layouts
- Open the default Shipment Layout

Sections to Add:

Section 1: Shipment Details

- Tracking Number
- Customer
- Shipment Type
- Weight
- Price
- Status

Section 2: Address Information

- Source Address
- Destination Address

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Arrange fields properly
- Click Save

Activity 3: Edit Page Layout for Delivery_Agent__c

Steps:

- Setup → Object Manager → Delivery_Agent__c
- Page Layouts
- Open default Delivery Agent Layout

Arrange Fields:

Agent Details

- Agent Name
- Email
- Phone
- Branch
- Status
- Joining Date

Additional Information

- Related Shipments

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 4: Edit Page Layout for Branch__c

Steps:

- Setup → Object Manager → Branch_c
- Page Layouts
- Open default Branch Layout

Sections:

Branch Information

- Branch Name
- Branch Code
- City
- State
- Manager Name
- Arrange fields
- Click Save

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

- Page Layouts
- Open default Delivery Update Layout

Sections:

Update Information

- Update Number
- Shipment
- Status
- Update Date
- Remarks
- Arrange properly
- Click **Save**

Activity 6: Edit Page Layout for Invoice__c

Steps:

- Setup → Object Manager → Invoice__c
- Page Layouts
- Open default Invoice Layout

Sections:

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Amount
- Invoice Date
- Payment Status
- Arrange fields as needed
- Click Save

Milestone 7 - Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Activity 1: Auto-Update Shipment Status When Delivery Update is Created

Scenario:

Whenever a Delivery_Update__c record is created, automatically update the related Shipment__c Status with the latest status from Delivery Update.

Part 1: Create the Record-Triggered Flow

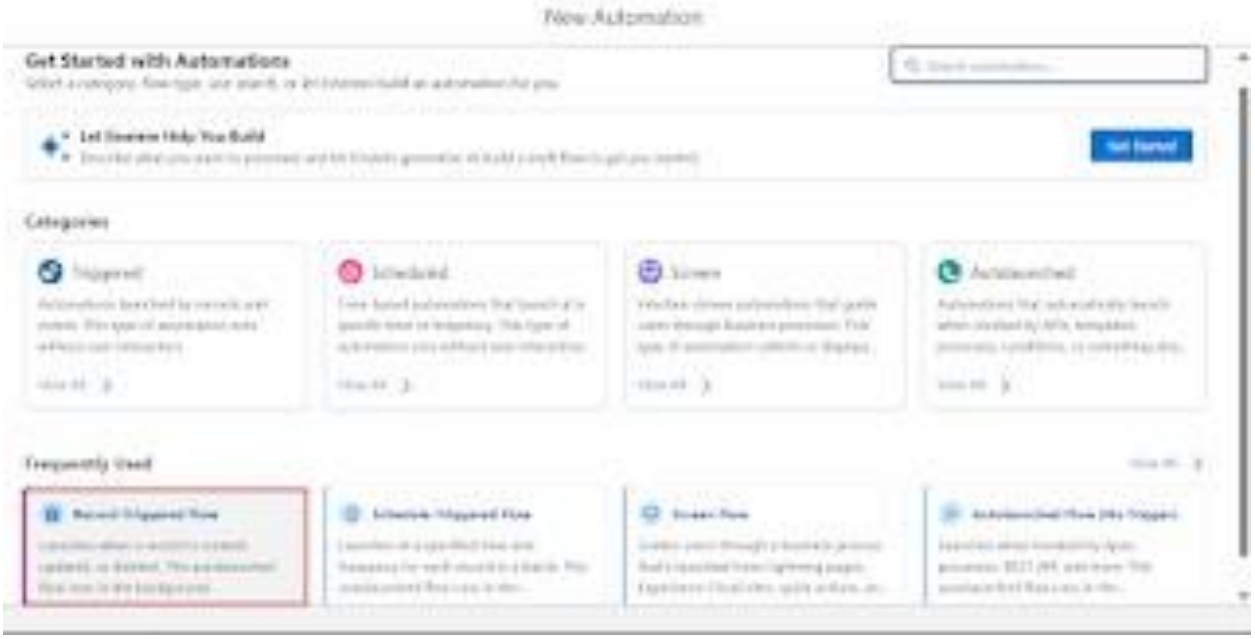
Step 1: Open Flow Builder

- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Flows
- Click Flows

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP



- Select Record-Triggered Flow
- Click Create



Step 2: Configure the Start (Trigger)

On the Start panel on the right:

- Object: select Delivery_Update__c
- Trigger the Flow When:
    - Select A record is created
- Condition Requirements:

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Click Done

Part 2: Update Related Shipment Record

Step 3: Add “Update Records” Element

- Click the + icon under the Start element
- Select Update Records

Configure the Update Records Element:

- Label: Update_Shipment_Status
- How to Find Records to Update:
    - Select Specify conditions to identify records

Set Filter Conditions:

- Object: Shipment__c
- Field: Id
- Operator: Equals
- Value: {!$Record.Shipment__c}

(This means update the Shipment linked to the Delivery Update)

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Status__c	{!$Record.Status__c}

(This copies status from Delivery Update to Shipment)

- Click Done

Part 3: Save and Activate

Step 4: Save Flow

- Click Save (top-right)
- Flow Label: Auto_Update_Shipment_Status
- API Name will auto-fill
- Click Save

Step 5: Activate Flow

- Click Activate
- Flow status should show Active

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



Milestone 8 - The Lightning App :

What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: Create a Lightning App

Create the Courier Management System App

1. From Setup, enter App Manager in the Quick Find and select App Manager.
2. Click New Lightning App.
    - Enter the Courier Management System as the App Name.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



4. Under App Options, leave the default selections and click Next.
5. Under Utility Items, leave as it is and click Next.
6. From Available Items, select the following and move them to Selected Items:
    - Customer
    - Shipment
    - Delivery Agent
    - Branch
    - Delivery Update
    - Invoice
    - Reports

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



9. Click Save & Finish.

Milestone 9- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types,

Login hours & Login IP ranges.

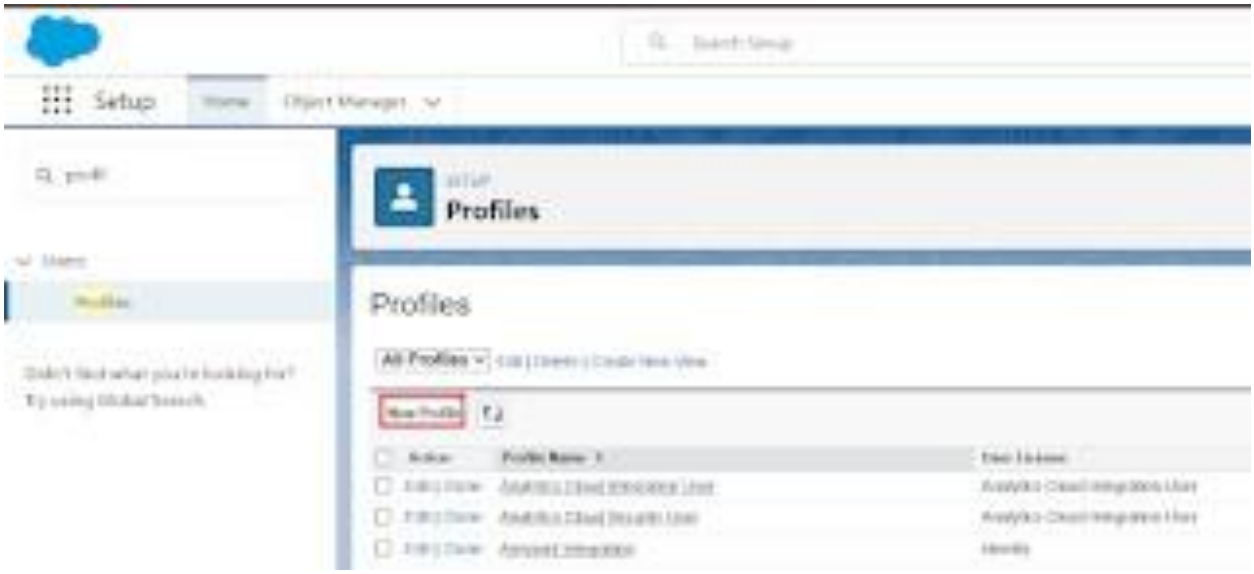
A profile can be assigned to many users, but users can be assigned a single profile at a time.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

2. Click Profiles.

3. Click New Profile.

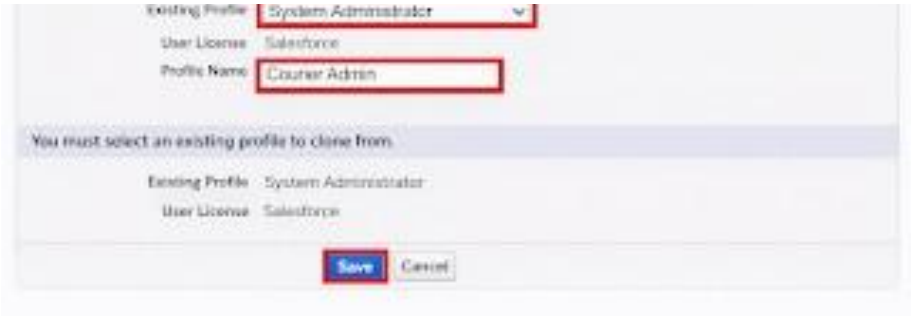


4. Select Existing Profile as: System Administrator.

5. Enter Profile Name: Courier Admin.

6. Click Save.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP



Why use System Administrator as Base?

It already has:

- Modify All Data
- Customize Application
- Full CRUD access on all objects

By cloning it, the **Courier Admin profile automatically inherits full access**.
You can later restrict permissions if required.

Activity 2: Create a Custom Profile – Branch Manager

Steps:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.
4. Select Existing Profile as: Standard Platform User.
5. Enter Profile Name: Branch Manager.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Under Assigned Apps / Custom App Settings, set the default app to:

Courier Management System App

Ensure Visible Tabs Include:

Set these tabs as Default On or Visible:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

Custom Object Permissions

Scroll to Custom Object Permissions and set access as follows:

Object	Permissions to Grant
Customer__c	Create, Read, Edit

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

Branch__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Invoice__c	Read



Session & Password Settings

Under Session Settings:

- Session Timeout: 2 Hours of Inactivity

Under Password Policies:

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 3: Create a Custom Profile – Delivery Agent

Steps:

1. Setup → Quick Find → Profiles
2. Click New Profile
3. Existing Profile: Standard Platform User
4. Profile Name: Delivery Agent
5. Click Save

App & Tab Access

Default App: Courier Management System

Visible Tabs:

- Shipments
- Delivery Updates

Object Permissions :

Object	Permissions
Shipment__c	Read, Edit
Delivery_Update__c	Create, Read, Edit

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Restrictions

- No access to Branch editing
- No delete permissions
- Limited reporting access

Click Save.

Activity 4: Create a Custom Profile – Customer Support

Steps:

- Setup → Profiles → New Profile
- Base Profile: Standard User
- Profile Name: Customer Support

Access Configuration

Tabs Visible:

- Customers
- Shipments
- Delivery Updates

Object Permissions :

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Shipment__c	Read, Edit
Delivery_Update__c	Read
Invoice__c	Read

Session Settings

- Session Timeout: 2 Hours
- Password Length: 8

Click Save

Final Profiles Created in Project :

Profile Name	Purpose
Courier Admin	Full system control
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status
Customer Support	Handle customers and tracking

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity 1: Creation of Courier Admin Role

Creation of Role:

1. From the Quick Find box, search for Roles and click on the Roles option.

2. Select Set Up Roles.



3. Click on Expand All and click Add Role under CEO.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



4. Give Label as “Courier Admin”.
    - The Role Name will auto-populate.
5. Click Save.

Activity 2: Creating Branch Manager Role Under Courier Admin Role

Create another role:

1. Create a role with the name “Branch Manager”.
2. Below the Courier Admin, click on Add Role.
3. Enter the label name as “Branch Manager” and the Role Name will automatically populate.
4. Click Save.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

1. Below the Branch Manager role, click on Add Role.
2. Enter Label as “Delivery Agent”.
    - Role Name will auto-populate.
3. Click Save.

Activity 4: Creating Customer Support Role Under Branch Manager

Steps:

1. Under the Branch Manager role, click on Add Role.
2. Enter Label as “Customer Support”.
3. Role Name will automatically populate.
4. Click Save.

Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking

This hierarchy ensures proper data sharing and security in the Courier Management System.

Milestone 11- User :

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.
NO THANKS USE THE APP

NOTE: As Salesforce license can only be used by 2 users at a time in Dev Org, so if you don't find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

Activity 1: To Create a User

Steps:

1. From Setup, enter Users in the Quick Find box, then select Users.

2. Click New User.



Fill in the Fields:

Field Name	Value
First Name	Enter first name (Example: Ravi)
Last Name	Enter last name (Example: Kumar)
Alias	Give an alias name
Email ID	Enter your personal email ID
Username	Must be in the format: text@text.text and must be unique
Nickname	Give a nickname
Role	Select **Courier Admin**

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 2: Creating Other Users

Repeat the same steps as **Activity 1** for creating additional users.

Use the following details:

User Type	Role	User License	Profile
Branch Manager	Branch Manager	Salesforce	Branch Manager
Delivery Agent	Delivery Agent	Salesforce Platform	Delivery Agent
Customer Support	Customer Support	Salesforce Platform	Customer Support

Example:

To create a Branch Manager User:

- Go to Setup → Users → New User
- Fill details like:

Field	Example Value
First Name	Arjun
Last Name	Reddy
Email	arjun@gmail.com
Username	arjun.manager@cms.com
Role	Branch Manager
Profile	Branch Manager

- Click Save

\

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Profile: Delivery Agent

To Create Customer Support User:

- Role: Customer Support
- Profile: Customer Support

Important Points

- Every user must have:
  - Unique username
  - Valid email
  - Assigned Role
  - Assigned Profile
- Profiles control permissions
- Roles control data visibility

Milestone 15 : Reports & Dashboards

Reports and Dashboards help analyze business data and track courier operations efficiently.

Activity 1: Create Report Types

Step 1: Open Report Types

1. Click Gear Icon → Select Setup.

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Primary Object: Shipment__c
- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 3: Create Customer Report Type

- Primary Object: Customer__c
- Label: Customer Summary Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 4: Create Agent Performance Report Type

- Primary Object: Delivery_Agent__c
- Label: Agent Performance Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

1. Click App Launcher (9 dots) → Search Reports
2. Click New Report
3. Choose Shipment Tracking Report
4. Click Continue

Step 2: Add Columns

Add these fields:

- Tracking Number
- Customer
- Status
- Expected Date
- Assigned Agent

Step 3: Add Filters

- Show Me: All Shipments
- Date Filter: This Month
- Status Filter: Select specific Status (Optional)

Step 4: Group & Summarize

- Group rows by Status__c

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

- Name: Shipment Status Summary
- Folder: CMS Reports

Report 2: Agent Performance Report

Step 1: Create Report

- New Report → Choose Agent Performance Report
- Click Continue

Step 2: Add Columns

- Agent Name
- Branch
- Number of Shipments
- Status

Step 3: Group Data

- Group by Delivery_Agent__c

Step 4: Save Report

- Name: Agent Performance Report
- Folder: CMS Reports
- Save & Run

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Add fields:

- Tracking Number
- Customer
- Price
- Invoice Status

Group by:

- Branch
- Month

Save as: Revenue by Branch Report

Activity 3: Create Dashboard

Step 1: Open Dashboards

1. Click App Launcher (9 dots)
2. Search Dashboards
3. Click New Dashboard

Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Activity 4: Add Dashboard Components

Component 1: Shipment Status Overview (Donut Chart)

1. Click **+** Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

Component 2: Agent Performance (Bar Chart)

- Select report: Agent Performance Report
- Chart Type: Bar Chart
- Click Add

Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Share Dashboard With:

- Courier Admin
- Branch Managers
- Customer Support Team

Final Layout :

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP



7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

9. Conclusion:

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP

Edit with the Docs app
Make tweaks, leave comments, and share with others to edit at the same time.

NO THANKS USE THE APP